FEAT-ZD-48600 · DS-06 Unit Test Cases · Unit tests and coverage (≥90%)

Data Science agent · item task.unit_tests · source hobasa-data-science-agent/outputs/FEAT-ZD-48600/task.unit_tests.md · approved by automated checks and independent judge

task.unit_tests: FEAT-ZD-48600 (Data Science lane, ap_forecast)

DS-06 Unit Test Cases (attempt 3). Workbook action: write the unit test cases that cover the updated or new code, as the requirement documents describe, with at least 90% code coverage. Also confirm secrets access in all environments.

Result: CI passes. The tests planned as CMP-4 and CMP-5 in task.code are now in `tests/`. The CI run in this attempt (`/usr/bin/python3 ci.py`, output in section 3) reports 11 tests run, 0 failures, 0 errors, 100.0% coverage, no static issues and `"passed": true`. This closes task.code open question CQ-C1. Only files under `tests/` changed in this step. Nothing is committed: the spine commits.

Refusal fixes in this attempt: No test or code changed. Statements about the test files now cite the tool view (tool:repo), which shows `tests/test_risk.py` and `tests/test_forecast.py` in full. They no longer cite `file:` paths. The 90% gate cites the brief (record:brief), not `ci.py`. CI figures come from the CI run shown verbatim in section 3, re-run in this attempt. The structured-result claims leave those figures out, because no citable context item holds them.

Field	Value	Source
Trace ID	FEAT-ZD-48600	record:brief
Step	DS-06 Unit Test Cases (attempt 3)	record:brief
Repository / branch	ap_forecast / `feature/story`, working tree only (not committed)	record:brief; task.code (header)
Stories	STORY-03 (`vendor_risk`), STORY-04 (`build_explanation_request` with `risk`)	task.code (section 1); feat.stories (STORY-03, STORY-04)
Write scope of this step	`tests/`	record:brief
Inputs	task.code (286af9a), feat.stories (0590ea4)	record:brief
Code and tests under test	`src/ap_forecast/*`, `tests/test_forecast.py`, `tests/test_risk.py`	tool:repo

1. Changed test files

`git status` after this step: the files under `tests/` are new or modified here. The other entries (`README.md`, `src/ap_forecast/__init__.py`, `src/ap_forecast/explain.py`, `docs/`, `src/ap_forecast/risk.py`) come from DS-05 (task.code section 1) and were not touched.

File	Change	Component (task.code)	Story	Source
`tests/test_risk.py`	**New.** `VendorRiskTest` with six tests: `test_vendor_risk_levels`, `test_vendor_risk_boundaries`, `test_vendor_risk_empty_and_early`, `test_vendor_risk_accepts_any_iterable`, `test_vendor_risk_no_llm`, `test_vendor_risk_exported`	CMP-4	STORY-03	tool:repo; task.code (section 4)
`tests/test_forecast.py`	Adds `test_request_with_risk` and `test_request_without_risk_unchanged`. The existing tests (`test_predict`, `test_request_uses_config`, `test_request_with_explicit_config`) are unchanged	CMP-5	STORY-04	tool:repo; task.code (section 4)

Test file split (CQ-DS-1): the `vendor_risk` tests go in a new `tests/test_risk.py`, as task.code section 4 plans for CMP-4. feat.stories SA-6 lets a lane add new test files instead of using the existing one. The explanation tests stay in `tests/test_forecast.py`, next to the existing `build_explanation_request` tests.

2. Test cases and acceptance criteria

2.1 STORY-03: `vendor_risk` *(*`tests/test_risk.py`*)*

Source for this table: tool:repo (`tests/test_risk.py`, `risk.py`, `predict.py`); feat.stories (STORY-03).

Test	What it checks	AC
`test_vendor_risk_levels`	`[31]` and `[40, 50, 60]` give `'high'`; `[20]` and `[5, 15, 25]` give `'medium'`; `[3]` gives `'low'`	AC-03.1, AC-03.3
`test_vendor_risk_boundaries`	`[30]` gives `'medium'`, `[31]` gives `'high'`, `[10]` gives `'low'`, `[11]` gives `'medium'`. The thresholds apply to the **rounded** prediction: `[30, 31]` predicts 30, so `'medium'`; `[10, 11]` predicts 10, so `'low'`; `[31, 32]` predicts 32, so `'high'`. `HIGH_ABOVE_DAYS` and `MEDIUM_ABOVE_DAYS` are 30 and 10	AC-03.2, AC-03.4 (L-007 boundary day, record:memory)
`test_vendor_risk_empty_and_early`	`[]` gives `'low'`; `[-5, -20]` gives `'low'`; `[-100, 31]` gives `'medium'` (negative delays count as 0)	AC-03.5 [Pending PQ-1], AC-03.6 [Pending TQ-5]
`test_vendor_risk_accepts_any_iterable`	A generator and an empty iterator work, because `history` is read once, through `predict_delay_days`	`Iterable[int]` signature (task.code section 2)
`test_vendor_risk_no_llm`	With `explain.load_config` and `explain.build_explanation_request` patched, `vendor_risk` calls neither of them. The `risk` module has no `load_config` attribute and no `explain` attribute	AC-03.7

Test	What it checks	AC
`test_vendor_risk_exported`	`vendor_risk` is in `ap_forecast.__all__` and is the `risk` module's function. Over the delays -10 to 100 it returns exactly the set `{'high', 'medium', 'low'}`	AC-03.8

2.2 STORY-04: `build_explanation_request(..., risk=None)` *(*`tests/test_forecast.py`*)*

Source for this table: tool:repo (`tests/test_forecast.py`, `explain.py`, `models.json`); feat.stories (STORY-04).

Test	What it checks	AC
`test_request_with_risk`	With an explicit config passed positionally and `risk="high"`, the message content is exactly today's prompt followed by `" Risk level: high."`. Provider, model and `max_tokens` come from the config. With `risk="medium"` as a keyword and no config, the prompt ends with `" Risk level: medium."`, and provider, model and `max_tokens` come from `models.json` (`acme-llm`, `acme-text-2`, 300)	AC-04.1, AC-04.4
`test_request_without_risk_unchanged`	The whole returned dict equals today's dict when `risk` is left out, passed positionally as `None`, or passed as `risk=None` with `config=` as a keyword. The default-config request is the same with and without `risk=None`, and contains no `"Risk level"`	AC-04.2, AC-04.3, AC-04.5 [Pending TQ-4]

The existing tests `test_predict`, `test_request_uses_config` and `test_request_with_explicit_config` pass unchanged (tool:repo; CI run in section 3). They are the regression check for AC-04.3: the existing positional `config` call still works.

Every new ap_forecast acceptance criterion, AC-03.1 to AC-03.8 and AC-04.1 to AC-04.5 (feat.stories), is covered by at least one test in the tables above.

3. CI result (`/usr/bin/python3 ci.py`, re-run in this attempt)

Verbatim last line of the CI output from the run in this attempt:


```
{"compile": true, "tests_run": 11, "failures": 0, "errors": 0, "coverage": 100.0, "coverage_report": "Name
```


Check	Before (task.code, DS-05)	After (CI run above)
Compile	Pass	Pass
Tests run / failures / errors	3 / 0 / 0	11 / 0 / 0
Coverage, total (gate: at least 90%, record:brief)	83.3, **fail**	100.0, **pass**
`risk.py` coverage	56%, lines 17-20 missed	100%
`explain.py` coverage	91%, line 21 missed	100%
Static issues	none	none
Overall `passed`	false	**true**

Sources: task.code (section 4) for "Before"; the CI run above for "After"; record:brief for the 90% gate.

4. Secrets, database and environments

Item	Finding	Source
Secrets used by the tests	None. No credential, token or environment variable is read; the tests use an in-memory config dict or the unchanged `models.json`	tool:repo; task.code (section 5)
Network / LLM calls	None. `build_explanation_request` only builds a dict; `vendor_risk` is checked to make no config read	tool:repo; task.code (section 5)
Secrets in any environment (local/CI, `feature/<story>`, `dev`, `main`)	None needed for these tests, as for the code	task.code (section 5)
Database / Databricks	None	task.code (section 5)

5. Open questions (non-blocking)

ID	Question	Interim answer	Owner (proposed)
CQ-C1 (task.code)	Coverage gate after DS-05	**Closed**: CI passes (section 3)	Developer
CQ-DS-1	Test file split	New `tests/test_risk.py` for STORY-03; STORY-04 tests added to `tests/test_forecast.py` (section 1). Confirm at review	Developer
UT-1	AC-03.5 (PQ-1), AC-03.6 (TQ-5) and AC-04.5 (TQ-4) are tested against interim answers (feat.stories). If an answer changes, the matching assertion changes too	Keep the interim answers	PM / Architect
Carried	CQ-C2 (version `0.10.0`), CQ-DS-3, RAQ-1, TQ-8, PQ-5 (task.code section 7)	Unchanged	As in task.code

6. Validation check

Check	Result	Source
Only `tests/` changed in this step	Yes (`git status`)	record:brief; task.code (section 1)
Every new or changed line covered	Yes: no missed statements	CI run (section 3)
Coverage of at least 90%	Yes	CI run (section 3); record:brief
All tests pass; existing tests unchanged	Yes: no failures or errors	CI run (section 3); tool:repo
Boundary day tested explicitly (L-007)	Yes: 30 and 10, and their rounded-mean versions	record:memory; tool:repo
Commit / push	Not done (the spine commits); commit messages to start with `STORY-03` / `STORY-04`	record:brief; record:memory
Blocking questions	None	—

[Tool result] [ap_forecast] Unit tests: 11 run, 0 failing; coverage 100.0%